Temperature and Humidity Monitoring System Using Arduino

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1. Introduction

Monitoring temperature and humidity in real time has applications in many fields such as agriculture, weather stations, home automation, industrial environments, and healthcare. This project demonstrates a basic yet functional embedded system that continuously measures and displays temperature and humidity values using an Arduino UNO microcontroller and a DHT11 sensor.

The DHT11 is a low-cost digital sensor that outputs calibrated digital signals for both temperature and relative humidity. It uses a single-wire serial interface and can be read easily by any microcontroller with a digital GPIO pin. The data is transmitted to a host computer and displayed on the Arduino IDE Serial Monitor.

2. Aim

To design and implement a temperature and humidity monitoring system using an Arduino UNO and a DHT11 sensor, and to display the measured values on the serial monitor at regular intervals.

3. Components Required

- Arduino UNO microcontroller board
- DHT11 Temperature and Humidity Sensor
- Connecting wires (jumper wires)
- USB cable (Type B, for Arduino-to-PC connection)
- Arduino IDE (installed on PC)
- DHT sensor library (installed in Arduino IDE)

4. Circuit Description and Pin Connections

The DHT11 sensor has four pins: VCC, Data, NC (not connected), and GND. The VCC pin is connected to the 5V supply of the Arduino. The GND pin is connected to the GND of the Arduino. The Data pin is connected to Digital Pin 2 of the Arduino. The Arduino UNO is powered via the USB cable connected to a laptop or PC.

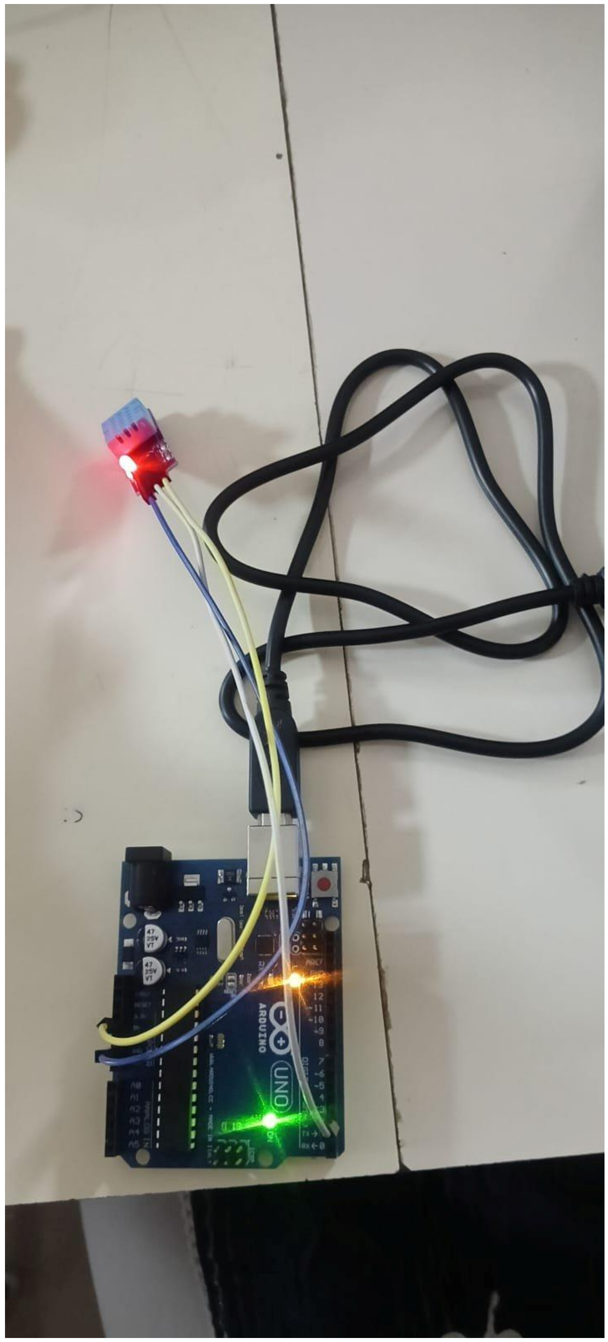

Figure 1: Hardware Setup — Arduino UNO with DHT11 Sensor

5. Working Principle

The DHT11 sensor uses a capacitive humidity sensing element and a thermistor to measure surrounding air conditions. After receiving a start signal from the microcontroller, the sensor sends back a 40-bit data packet (8-bit humidity integer, 8-bit humidity decimal, 8-bit temperature integer, 8-bit temperature decimal, and 8-bit checksum) using a single-wire protocol.

The Arduino, using the DHT library, decodes this data and stores it as floating-point values for temperature (in degrees Celsius) and relative humidity (in percentage). These values are then printed to the Serial Monitor every 2 seconds using the Serial.print() and Serial.println() functions.

If the sensor fails to respond or returns invalid (NaN) data, the code prints an error message to the Serial Monitor, helping in debugging hardware issues.

6. Arduino Code

```
#include <DHT.h>

#define DHTPIN 2      // Data pin connected to Arduino pin 2
#define DHTTYPE DHT11 // Change to DHT22 if using DHT22

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  dht.begin();
  Serial.println("Temperature and Humidity Monitoring");
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
  }

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.print(" °C");
  Serial.print("   Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");
  delay(2000);
}
```

7. Observations

The Serial Monitor output was recorded for ten consecutive readings taken at 2-second intervals. The observed values are tabulated below:

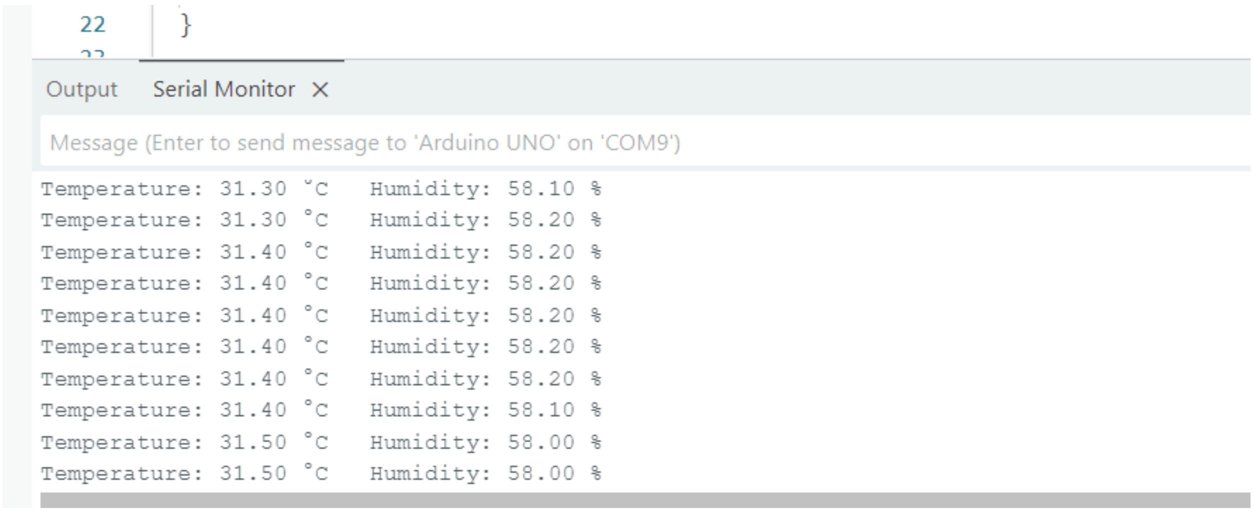

Figure 2: Serial Monitor Output

8. Results and Discussion

The system successfully read and printed temperature and humidity values from the DHT11 sensor at 2-second intervals without any read failures. From the observations, the temperature was found to be in the range of 31.30°C to 31.50°C, and the humidity was between 58.00% and 58.20%. These values are consistent with typical indoor room conditions and indicate that the sensor is working correctly.

The slight variation in temperature across readings (a rise from 31.30°C to 31.50°C over 10 readings) may be due to sensor self-heating or minor ambient temperature changes during the measurement period. The humidity shows a slight downward trend from 58.20% to 58.00%, which is within the DHT11 measurement accuracy of ±5% RH.

9. Conclusion

This project successfully demonstrates the interfacing of a DHT11 sensor with an Arduino UNO to monitor environmental temperature and humidity in real time. The system reads sensor data every 2 seconds and displays it accurately on the Serial Monitor. The implementation is simple, low-cost, and reliable, making it suitable as a foundational module for more advanced applications such as weather stations, greenhouse monitoring, HVAC control systems, and IoT-based smart environment monitoring.

Future improvements can include integrating an LCD or OLED display for standalone operation, adding data logging to an SD card or cloud platform, and incorporating threshold-based alerts using buzzers or LEDs.